

Problem Set 2

Probability and Statistical Inference

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1. A continuous random variable X has density

$$f_X(x) = \begin{cases} c(1+x), & -1 \leq x < 0, \\ c(1-x/2), & 0 \leq x \leq 2, \\ 0, & \text{otherwise,} \end{cases}$$

where c is a constant.

- (a) Find c and briefly explain why the resulting function is a valid density.
 - (b) Derive the CDF $F_X(x)$ over the whole real line. Check that your expressions agree at $x = 0$ and that the CDF reaches one at the upper end of the support.
 - (c) Sketch the density and the CDF. Label the support and the values at $x = -1, 0, 2$.
 - (d) Find $\Pr(-1/2 < X \leq 1)$.
 - (e) Calculate $\mathbb{E}[X]$ and $\text{Var}(X)$.
 - (f) The density has height $f_X(0) = 2/3$. What is $\Pr(X = 0)$? Explain why these two quantities are different.
2. Let $X \sim U[-2, 1]$ and define $Y = X^2$. The transformation is not one-to-one on the support of X : different values of X can produce the same value of Y .
- (a) State the support of Y . For each possible value y , identify which of the roots $x = -\sqrt{y}$ and $x = \sqrt{y}$ lie in the support of X . Explain why the answer changes at $y = 1$.
 - (b) Use the CDF method to derive $F_Y(y)$ for all $y \in \mathbb{R}$.
 - (c) Hence derive the density $f_Y(y)$. Verify that it integrates to one.
 - (d) Calculate $\Pr(1/4 \leq Y \leq 9/4)$.
 - (e) Calculate $\mathbb{E}[Y]$ using your density and verify the result directly from $Y = X^2$.
 - (f) A student applies the lecture's density-transformation formula using only the inverse $x = \sqrt{y}$. Describe precisely which inverse branch is missed and for which values of y .
3. The joint distribution of discrete random variables $X \in \{0, 1\}$ and $Y \in \{0, 1, 2\}$ is shown below.

	$X = 0$	$X = 1$	$\Pr(Y = y)$
$Y = 0$	1/4	0	
$Y = 1$	1/4	1/4	
$Y = 2$	0	1/4	
$\Pr(X = x)$			1

- (a) Complete the marginal probabilities in the table.
- (b) Find the conditional distribution of Y given $X = 0$ and the conditional distribution of Y given $X = 1$.
- (c) For each $x \in \{0, 1\}$, calculate $\mathbb{E}[Y \mid X = x]$, $\mathbb{E}[Y^2 \mid X = x]$, and $\text{Var}(Y \mid X = x)$.

- (d) Calculate $\mathbb{E}[Y]$ directly from its marginal distribution and verify the law of iterated expectations, $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}(Y | X)]$.
- (e) Calculate $\text{Var}(Y)$ directly and then verify the law of total variance,

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X]).$$

Interpret the two terms as within- X and between- X variation.

- (f) Are X and Y independent? Are they uncorrelated? Justify both answers.
4. Suppose that (X, Y) has joint density

$$f_{X,Y}(x, y) = \begin{cases} cy, & 0 < x < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Draw the support of (X, Y) and find c .
- (b) Derive the marginal densities $f_X(x)$ and $f_Y(y)$, stating their supports.
- (c) Derive $f_{X|Y}(x | y)$ and $\mathbb{E}[X | Y = y]$ for $0 < y < 2$. Give an intuitive description of this conditional distribution.
- (d) On a second sketch of the support, shade the event $X + Y \leq 2$. Explain why an integral over this event must be split at $y = 1$.
- (e) Calculate $\Pr(X + Y \leq 2)$.
- (f) Are X and Y independent? Give one argument based on either the support or the conditional density.

5. This question studies the chain

$$X \perp\!\!\!\perp Y \implies Y \text{ is mean independent of } X \implies \text{Cov}(X, Y) = 0.$$

Assume throughout that the expectations used below exist.

- (a) Prove both implications in the displayed chain.
 - (b) Let X be equally likely to be 0 or 1. Conditional on $X = 0$, let Y be equally likely to be -2 or 2 ; conditional on $X = 1$, let Y be equally likely to be -1 or 1 . Show that Y is mean independent of X , but that X and Y are not independent.
 - (c) Let $X \sim U[-\pi, \pi]$ and set $Y = \cos X$. Show that X and Y are uncorrelated but that Y is not mean independent of X .
 - (d) Explain how parts (b) and (c) establish that neither reverse implication is valid in general.
6. A university recruits 200 volunteers for a study of an online revision workshop. Within this group, 100 students are randomly assigned to the workshop ($D = 1$) and 100 to the control group ($D = 0$). Let Y be the end-of-term score. Write the outcome in terms of its two group means:

$$Y = \alpha + \beta D + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D] = 0.$$

The realized sample means are $\bar{Y}_1 = 76$ and $\bar{Y}_0 = 70$.

- (a) Show that $\alpha = \mathbb{E}[Y \mid D = 0]$ and $\beta = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]$.
- (b) Estimate each population group mean by its sample group mean. Use part (a) to obtain $\hat{\alpha}$ and $\hat{\beta}$.
- (c) Using potential-outcome notation, state the random-assignment condition that allows a causal interpretation of the difference in means. For which population does the experiment identify an average causal effect directly?
- (d) In this description of the group means, interpret $\mathbb{E}[\varepsilon \mid D] = 0$ and show that it implies $\mathbb{E}[D\varepsilon] = 0$.
- (e) Does random assignment make the 200 volunteers representative of all university students? Explain the different roles of random assignment and random sampling.
- (f) Now suppose that the university took a random sample of students, but each student chose whether to attend the workshop. What would random sampling achieve? What additional problem would prevent an automatic causal interpretation of $\hat{\beta}$?